

- **Provision and management:** This entails managing millions of distributed IoT devices and gateways with added security.
 - **Data flow:** Here, data flow management is simplified, with intelligence and analytics at the edge.
 - **Analytics:** This provides a comprehensive, centralised, advanced analytics and data management platform with the ability to build or refine machine learning models and push these to the edge.
 - **Security:** This includes pre-integrated security and manageability across devices, as well as secure access and authentication for data that is in motion and at rest.
 - **Utilisation:** This comprises application development and deployment, and integration services.
- Figure 1 depicts the open source IoT ecosystem.

Open source development platforms and frameworks

IoT platforms offer the support software that connects everything in an IoT system. They facilitate communication, data flow, device management and the functionality of

applications. A few leading open source IoT development platforms for IoT based applications are listed below.

Kaa: Kaa is a flexible, multi-purpose platform for developing end-to-end IoT solutions for connected applications. It provides better communication, control and interoperation capabilities in connected products and smart devices. It is primarily based on flexible microservices and easily adapts to almost any need and application.

Sitewhere: This platform facilitates ingestion, storage, processing and integration of device data. It provides connectivity of devices using different protocols like MQTT, SMQP, STOMP, etc; devices can even be added through self-registration, REST services or in batches. The platform is based on microservices architecture, and is designed for reliability, high throughput, low latency and stability.

DeviceHive: This facilitates the connectivity of devices to the cloud within a short interval of time, i.e., just a few minutes, to allow streaming of data and command traversal. The platform is distributed under the Apache 2.0 licence and is highly

scalable via containerisation. It supports Docker and Kubernetes.

Zetta: This API based IoT platform built on Node.js is regarded as a complete toolkit to generate HTTP APIs for devices. Zetta combines REST APIs, WebSockets and reactive programming to build data-intensive and real-time applications.

Thingsboard.io: This is used for data collection, processing, visualisation and device management. It supports all standard IoT protocols like CoAP, MQTT and HTTP, as well as cloud and on-premise deployments.

Other popular open source IoT platforms are DSA, Thinger.io, Mainflux, Thingspeak, etc.

Open source middleware

Adoption of open source middleware enhances interoperability with other enterprise applications because of reuse of recommended software stacks, libraries and components. Open source middleware increases the productivity and quality of applications, speeds up the time to market, and reduces development risks. The following are a few top-of-the-line open source middleware platforms for IoT based applications.

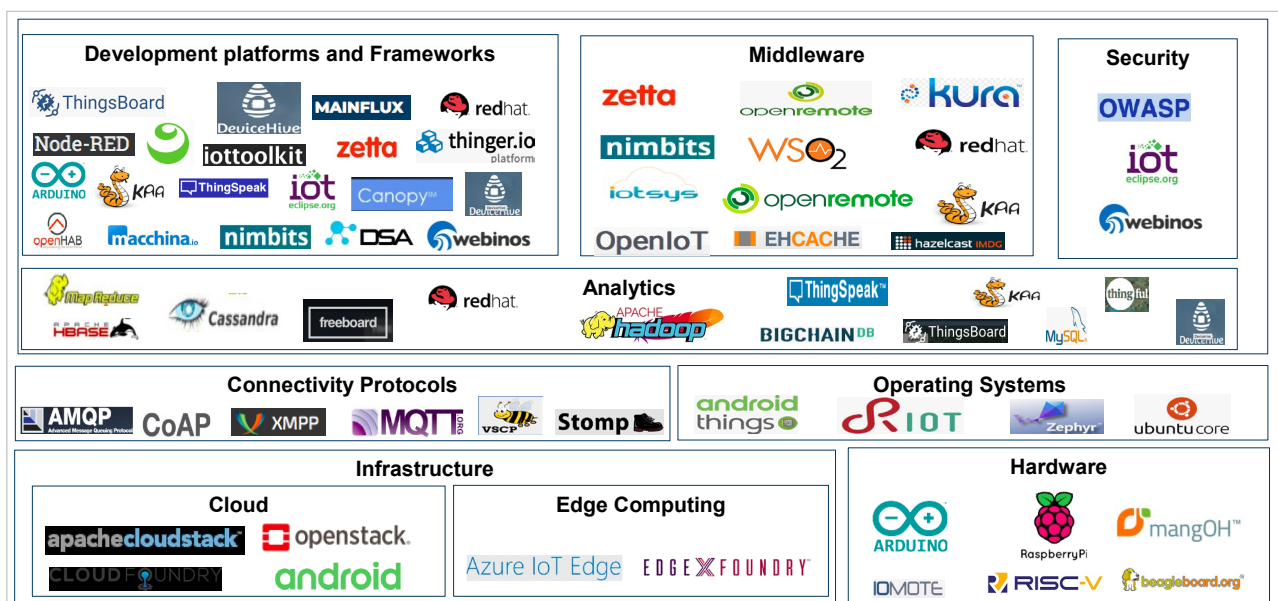


Figure 1: Open source IoT ecosystem